

JavaScript Set Logic

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The union() Method

`union()` creates a new Set containing all elements from both Sets:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Set Logic</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The union() Method</h4>
<p id="demo"></p>

<script>
const setA = new Set(["a", "b", "c"]);
const setB = new Set(["c", "d", "e"]);

let text = "Set A: " + [...setA] + "<br>";
text += "Set B: " + [...setB] + "<br><br>";

if (setA.union) {
  const result = setA.union(setB);
  text += "union(): " + [...result] + "<br>";
  text += "All elements from both sets (no duplicates)";
} else {
  text += "union() is not supported in this browser yet.";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

The intersection() Method

`intersection()` creates a new Set with elements in BOTH Sets:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Set Logic</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The intersection() Method</h4>
<p id="demo"></p>

<script>
const setA = new Set(["a", "b", "c"]);
const setB = new Set(["c", "d", "e"]);

let text = "Set A: " + [...setA] + "<br>";
text += "Set B: " + [...setB] + "<br><br>";

if (setA.intersection) {
  const result = setA.intersection(setB);
  text += "intersection(): " + [...result] + "<br>";
  text += "Only elements present in BOTH sets";
} else {
  text += "intersection() is not supported in this browser yet.";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

The difference() Method

`difference()` creates a new Set with elements in A but NOT in B:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Set Logic</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The difference() Method</h4>
<p id="demo"></p>

<script>
const setA = new Set(["a", "b", "c"]);
const setB = new Set(["c", "d", "e"]);

let text = "Set A: " + [...setA] + "<br>";
text += "Set B: " + [...setB] + "<br><br>";

if (setA.difference) {
  const result = setA.difference(setB);
  text += "difference(): " + [...result] + "<br>";
  text += "Elements in A but NOT in B";
} else {
  text += "difference() is not supported in this browser yet.";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

Example 3

Result [View Example](#)

The symmetricDifference() Method

`symmetricDifference()` creates a new Set with elements in either Set, but NOT in both:

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Set Logic</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The symmetricDifference() Method</h4>
<p id="demo"></p>

<script>
const setA = new Set(["a", "b", "c"]);
const setB = new Set(["c", "d", "e"]);

let text = "Set A: " + [...setA] + "<br>";
text += "Set B: " + [...setB] + "<br><br>";

if (setA.symmetricDifference) {
  const result = setA.symmetricDifference(setB);
  text += "symmetricDifference(): " + [...result] + "<br>";
  text += "Elements in either set but NOT in both";
} else {
  text += "symmetricDifference() is not supported in this browser yet.";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>

```

Example 4

Result [View Example](#)

isSubsetOf(), isSupersetOf(), isDisjointFrom()

These methods check the relationships between two Sets:

- `isSubsetOf()` - true if every element of A is in B
- `isSupersetOf()` - true if every element of B is in A
- `isDisjointFrom()` - true if A and B have no common elements

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Set Logic</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>isSubsetOf(), isSupersetOf(), isDisjointFrom()</h4>
<p id="demo"></p>

<script>
const setA = new Set(["a", "b"]);
const setB = new Set(["a", "b", "c", "d"]);
const setC = new Set(["x", "y"]);

let text = "Set A: " + [...setA] + "<br>";
text += "Set B: " + [...setB] + "<br>";
text += "Set C: " + [...setC] + "<br><br>";

if (setA.isSubsetOf) {
  text += "A.isSubsetOf(B): " + setA.isSubsetOf(setB) + "<br>";
  text += "B.isSupersetOf(A): " + setB.isSupersetOf(setA) + "<br>";
  text += "A.isDisjointFrom(C): " + setA.isDisjointFrom(setC) + "<br><br>";
  text += "A is subset of B ✓<br>";
  text += "B is superset of A ✓<br>";
  text += "A and C are disjoint ✓";
} else {
  text += "These methods are not supported in this browser yet.";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>

```

Example 5

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Set Logic](#)